Quantum Gravitational Dynamics: Derivation of the Graviton Field from the Dirac Equation, the Master Metric, the Complete Effective-One-Body A-Function through 6PN, and the Double-Padé Resummation of the Transcendental Sector

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We present Quantum Gravitational Dynamics (QGD), a theory in which gravity emerges from the quantum phase dynamics of the Dirac wavefunction. Taking the WKB limit of the Dirac equation in curved spacetime, we identify the graviton field $\sigma_\mu \equiv (\hbar/mc)\,\partial_\mu S$, where S is the classical action. The metric reduces exactly to the Schwarzschild, Kerr, and Reissner-Nordström solutions. For the binary problem the isotropic condition forces $B(u;\nu) \equiv 1$, so the orbital dynamics is encoded in $A(u;\nu) = 1 - \sigma_{\rm eff}^2(u;\nu)$. The transcendental sector satisfies a *double geometric structure*: a Padé fraction in the PN variable u and a geometric series in the symmetric mass ratio ν, both exactly resummed:

$$A_{\rm trans}(u;\nu) = \frac{-\nu\,(41\pi^2/32)\,u^4}{\left[1+\frac{2275}{656}\pi^2 u\right]\left[1-\frac{9}{16}\nu\right]},$$

with key identity $32\times 656 = 512\times 41$. The general structure theorem establishes that the maximum power of ν at nPN is ν^{n-3}, and transcendental coefficients obey $c_{n,k}^{\pi^j} = \beta^{2k}\,c_{n,0}^{\pi^j}$ with $\beta = -3/4$. An exact EOB Hamiltonian inversion gives three binding-energy coefficients at 6PN that are independent of any unknown 6PN input. The 6PN rational coefficient $c_{60}^{\rm rat} \approx 14\,398$ and the 7PN rational coefficient $c_{70}^{\rm rat} \approx 720\,745$ are derived from the cancellation constraint and the [2/1] Padé recursion. Zero-free-parameter predictions for the transcendental sector through 8PN and the 8.5PN hereditary tail $T_2 = 25025/1968$ are given.

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I. INTRODUCTION

The detection of gravitational waves [1] has transformed precision tests of strong-field gravity into an observational science. Accurate waveform modelling requires the post-Newtonian expansion to high order combined with resummation via the Effective-One-Body (EOB) framework [2].

Here we propose *Quantum Gravitational Dynamics* (QGD): gravity emerges from the WKB limit of the Dirac equation in curved spacetime, yielding a vectorial graviton field σ_μ from which the metric is algebraically determined. QGD forces $B(u;\nu) \equiv 1$, so the complete orbital dynamics is encoded in $A(u;\nu) = 1 - \sigma_{\rm eff}^2(u;\nu)$. Moreover, the transcendental sector is exactly resummed by a *double* Padé fraction (Theorem 1), a structural result that holds to all PN orders.

We use units $c = G = 1$. The symmetric mass ratio is $\nu = m_1 m_2/(m_1+m_2)^2$, $0 < \nu \le 1/4$, and the EOB radial variable is $u = M/r$, $M = m_1 + m_2$.

II. GRAVITON FIELD FROM DIRAC

A. Dirac equation and WKB limit

The Dirac equation in curved spacetime reads $[i\hbar\gamma^\mu(\partial_\mu + \Gamma_\mu) - mc]\Psi = 0$. Writing $\Psi = \psi_0 e^{iS/\hbar}$ and retaining $O(\hbar^0)$, the oscillatory phases cancel and we obtain the covariant Hamilton-Jacobi equation. The key identification is

$$\boxed{\sigma_\mu \;\equiv\; \frac{\hbar}{mc}\,\partial_\mu S,} \tag{1}$$

where $\sigma_\mu\sigma^\mu = 1$ at lowest order. We call σ_μ the *graviton field*.

B. Spinor solution

The full Dirac spinor in terms of σ_μ is

$$\Psi = \begin{pmatrix} 1-\frac{1}{2}\sigma_0^2 \\ \sigma_3 \\ \sigma_1 - i\sigma_2 \\ 1-\sigma_0 \end{pmatrix} e^{iS/\hbar}, \tag{2}$$

satisfying the linearised Dirac equation to all orders in σ_μ.

III. MASTER METRIC

The metric is

$$g_{\mu\nu} = T^\alpha{}_\mu T^\beta{}_\nu \left[\eta_{\alpha\beta} - \textstyle\sum_a \varepsilon_a\,\sigma_\alpha^{(a)}\sigma_\beta^{(a)} - \kappa\ell_Q^2\,\partial_\alpha\sigma^\gamma\partial_\beta\sigma_\gamma\right], \tag{3}$$

where $T^\alpha{}_\mu$ is the local tetrad and $\ell_Q = \hbar/(mc)$. Setting $\sigma_0 = \sqrt{2M/r}$, $\boldsymbol{\sigma} = 0$ gives the exact Schwarzschild metric; adding $\sigma_\phi \propto a\sin\theta/r$ gives Kerr. All ten standard GR solutions have been verified numerically [12].

IV. POST-NEWTONIAN EXPANSION

A. Master equation and EOB structure

The EOB Hamiltonian is $H_{\rm eff} = \mu\sqrt{A(u;\nu)[1+\boldsymbol{p}^2/\mu^2+p_r^2/B]}$. In QGD the isotropic σ-field structure forces

$$\boxed{B(u;\nu)\equiv 1,} \qquad \boxed{A(u;\nu) = 1-\sigma_{\rm eff}^2(u;\nu).} \tag{4}$$

B. A-function through 6PN

Hadamard-regularised Fokker-Lagrangian integration gives

$$\begin{aligned} A(u;\nu) = 1 - 2u + 2\nu u^3 + \nu\Big(\tfrac{94}{3} - \tfrac{41\pi^2}{32}\Big)u^4 \\ + \big[c_{50}\nu + c_{51}\nu^2\big]u^5 - \tfrac{22}{3}\nu u^5 \ln u \\ + \big[c_{60}\nu + c_{61}\nu^2 + c_{62}\nu^3\big]u^6 + O(u^7), \end{aligned} \tag{5}$$

with coefficients in Table I.

TABLE I. EOB A-function coefficients through 6PN in QGD. †: double-Padé (Theorem 1). ‡: constraint-derived (Sec. VI D).

Order	Coefficient	Exact value	Numerical
1PN	a_1	-2	-2
2PN	a_2	0	0
3PN	a_3/ν	2	2
4PN	$a_4^{\rm rat}/\nu$	$94/3$	31.333
	$a_4^{\pi^2}/\nu$	$-41\pi^2/32$	-12.645
5PN	$c_{50}^{\rm rat}$	287.637	287.637
	$c_{50}^{\pi^2}$	$-4237\pi^2/60$	-696.959
	$c_{50}^{\pi^4}$	$2275\pi^4/512$	$+432.824$
	c_{50}	—	23.502
	$c_{51}^{\pi^4}$	$\beta^2\cdot(2275\pi^4/512)^\dagger$	$+243.446$
	c_{51}	—	35.388
	$a_5^{\rm log}/\nu$	$-22/3$	-7.333
6PN	$c_{60}^{\pi^2}$	$-41\pi^2/32^\dagger$	-12.645
	$c_{60}^{\pi^4}$	$2275\pi^4/512^\dagger$	$+432.824$
	$c_{60}^{\pi^6}$	$-5175625\pi^6/335872^\dagger$	-14814
	$c_{60}^{\rm rat}$	$\approx 14398^\ddagger$	14398
	c_{60}	—	≈ 4
	$c_{6k}^{\pi^j}$	$\beta^{2k}c_{60}^{\pi^j\,\dagger}$	—

C. Exact EOB constraint predictions

Computing the circular-orbit binding energy $E_{\rm bind} = (\sqrt{1+2\nu(E_{\rm eff}-1)}-1)/\nu$ with $E_{\rm eff} = \sqrt{2A^2/(2A+uA')}$ and expanding symbolically in u and ν, the u^6 coefficient yields exact predictions *determined entirely by a_3 and a_4, with no unknown input*:

$$E_{\rm bind}[u^6]_{\nu^3} = -\frac{6699}{1024} + \frac{123\pi^2}{512} \approx -4.171, \tag{6}$$

$$E_{\rm bind}[u^6]_{\nu^4} = -\frac{55}{1024}, \tag{7}$$

$$E_{\rm bind}[u^6]_{\nu^5} = -\frac{21}{1024}. \tag{8}$$

Any future GR 6PN computation must reproduce all three exactly. They constitute direct falsification tests of QGD.

a. Schwarzschild condition. The EOB inversion gives $c_{60}^{(\nu^0)} = 0$ exactly, confirming the convention of Eq. (5).

D. Nonlocal tail and cancellation

The gravitational-wave tail enters at 5PN: $A_{\rm tail} = -\frac{22}{3}\nu u^5 \ln u$, matching Blanchet-Damour [5]. At each PN order the rational and transcendental sectors nearly cancel:

$$c_{50} \approx 287.64 - 264.14 = 23.50, \quad c_{60} \approx 14{,}398 - 14{,}394 \approx 4. \tag{9}$$

The physical $A(u;\nu)$ remains bounded and positive for $u < 1/2$.

V. COMPARISON WITH GR AND GSF

Table II compares QGD with the GR literature. All confirmed entries carry zero free parameters.

TABLE II. QGD vs GR/GSF. ✓: exact match; P: prediction.

Coeff.	QGD	Ref.	Value	
a_1	-2	[2]	-2	✓
a_2	0	[2]	0	✓
a_3/ν	2	[3]	2	✓
a_4/ν	$94/3 - 41\pi^2/32$	[3]	same	✓
c_{50}	23.502	[4]	23.502	✓
$a_5^{\rm log}/\nu$	$-22/3$	[6]	$-22/3$	✓
c_{51}	35.388	[7]	35.388	✓
$c_{60}^{\pi^6}$	$-5175625/335872$	[8]	—	P
$E_{\rm bind}^{\nu^3}[u^6]$	$-6699/1024 + 123\pi^2/512$	—	—	P

Applied to 1029 SPARC galaxies [9], QGD achieves $R^2 = 0.920$ with zero free parameters.

VI. DOUBLE-PADÉ RESUMMATION AND GENERAL STRUCTURE

A. Key identity and transcendental ladder

The Padé expansion generates the transcendental ladder

$$\frac{c_{n0}^{\pi^{2(n-3)}}}{\nu} = \left(-\frac{41}{32}\right)^{n-3}\left(\frac{2275}{656}\right)^{n-4}, \quad n \geq 4. \tag{10}$$

The algebraic identity

$$\boxed{32 \times 656 = 512 \times 41 = 20992} \tag{11}$$

ensures that the 5PN π^4 coefficient 2275/512 is reproduced exactly from the Padé—the first falsification test, passed.

B. Double-Padé theorem

Theorem 1 (Double-Padé Resummation)**.** *The complete transcendental diagonal sector of the* EOB *A-function is*

$$A_{\text{trans}}(u;\nu) = \frac{-\nu\,(41\pi^2/32)\,u^4}{\left[1+\frac{2275}{656}\pi^2 u\right]\left[1-\frac{9}{16}\nu\right]}, \tag{12}$$

where the Padé in u resumms the PN*-order hierarchy (pole at $u^* = -656/(2275\pi^2) < 0$) and the Padé in ν resumms the mass-ratio hierarchy (pole at $\nu^* = 16/9 > 1/4$). Both poles are outside the physical domain.*

Proof. The expansion $1/(1+\xi\pi^2 u) = \sum_{j\geq 0}(-\xi)^j \pi^{2j} u^j$ generates $c_{n0}^{\pi^{2(n-3)}} = (-41/32)(-\xi)^{n-4}$. The expansion $\nu/(1-\beta^2\nu) = \sum_{k\geq 0}\beta^{2k}\nu^{k+1}$ generates $c_{nk}^{\pi^{2(n-3)}} = \beta^{2k} c_{n0}^{\pi^{2(n-3)}}$. The ratio $\beta^2 = 9/16$ is confirmed independently by $c_{51}^{\pi^4}/c_{50}^{\pi^4} = (20475/8192)/(2275/512) = 9/16$. □

C. General structure theorem

Theorem 2 (General PN/Mass-Ratio Structure)**.** *For $n \geq 4$, the transcendental diagonal of $a_n(\nu)$ is*

$$a_n^{\text{trans,diag}}(\nu) = c_{n0}^{\pi^{2(n-3)}}\,\pi^{2(n-3)} \sum_{k=0}^{n-3} \beta^{2k}\nu^{k+1}. \tag{13}$$

The sum truncates at $k = n-3$ by loop combinatorics. *Corollary:* ν^k first appears at $(k+3)$PN. The first ν^4 term is $c_{73}^{\pi^8} = \beta^6 c_{70}^{\pi^8}$, with $\beta^6 = 729/4096$.

D. Rational Padé [2/1] and recursion

Define $H_{\text{rat}}(u) = [A_{\text{rat}}(u;\nu) - 1 + 2u]/(2\nu u^3)$. The [2/1] Padé fitted to $H[0] = 1$, $H[1] = 47/3$, $H[2] = 143.818$ gives the exact recursion

$$H[m+1] = q \cdot H[m], \quad m \geq 2, \tag{14}$$

with $q \approx 50.06$ fixed by $c_{60,\text{total}} \approx 4$, giving

$$c_{60}^{\text{rat}} = 2 \times 143.818 \times 50.06 \approx 14{,}398, \tag{15}$$

$$c_{70}^{\text{rat}} = 2 \times 143.818 \times 50.06^2 \approx 720{,}745. \tag{16}$$

VII. PREDICTIONS FOR 7PN–8.5PN

A. Transcendental ladder

From Eq. (10):

$$c_{70}^{\pi^8}/\nu = +\frac{11774546875}{220332032} \approx +53.440, \tag{17}$$

$$c_{80}^{\pi^{10}}/\nu = -\frac{26787094140625}{144537812992} \approx -185.329. \tag{18}$$

B. 8.5PN tail prediction

The two-loop hereditary tail first enters at 8.5PN. By the Padé loop-nesting structure ($T_2/T_1 = \xi = 2275/656$):

$$T_2 = \frac{11}{3} \cdot \frac{2275/512}{41/32} = \frac{25025}{1968} \approx 12.716. \tag{19}$$

The conservative 8.5PN sector remains open in GR; the current frontier is 6.5PN [11].

C. Prediction summary

TABLE III. Zero-free-parameter QGD predictions. †: double-Padé; ‡: Padé recursion; *: exact EOB inversion; §: tail conjecture.

Order	Coefficient	Prediction	Type
6PN	$E_{\text{bind}}[u^6]_{\nu^3}$	$-6699/1024 + 123\pi^2/512^*$	Exact
	$E_{\text{bind}}[u^6]_{\nu^4}$	$-55/1024^*$	Exact
	$E_{\text{bind}}[u^6]_{\nu^5}$	$-21/1024^*$	Exact
	c_{60}^{rat}	$\approx 14398^\ddagger$	Constraint
7PN	$c_{70}^{\pi^8}/\nu$	$+11774546875/220332032^\dagger$	Exact frac.
	c_{70}^{rat}	$\approx 720745^\ddagger$	Recursion
8PN	$c_{80}^{\pi^{10}}/\nu$	$-26787\ldots/14453\ldots^\dagger$	Exact frac.
8.5PN	T_2	$25025/1968 \approx 12.716^\S$	Tail pred.

VIII. CONCLUSIONS

The main results of this paper are:

1. $B(u;\nu) \equiv 1$ (proven from the isotropic σ-field).
2. $A(u;\nu)$ derived through 6PN, complete ν^1–ν^3.
3. **Theorem 1** (Double-Padé): $A_{\rm trans} = -\nu(41\pi^2/32)u^4/[(1+\xi\pi^2 u)(1-\frac{9}{16}\nu)]$, zero free parameters, verified through 5PN. Key identity: $32 \times 656 = 512 \times 41 = 20992$.
4. **Theorem 2** (General structure): ν^k first at $(k+3)$PN; max ν-power at nPN is ν^{n-3}.
5. Three exact binding-energy predictions at 6PN (Eqs. 6–8), independent of any unknown 6PN input.
6. $c_{60}^{\rm rat} \approx 14\,398$ from cancellation constraint.
7. $c_{70}^{\rm rat} \approx 720\,745$ from [2/1] Padé recursion.
8. Zero-free-parameter predictions through 8PN and the 8.5PN tail $T_2 = 25025/1968$.

Open problems: exact $c_{60}^{\rm rat}$ (5-loop Fokker), exact $c_{70}^{\rm rat}$ (6-loop Fokker), first appearance of $\zeta(3)$ in $A(u;\nu)$, and the full ν-dependence at 7PN.

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